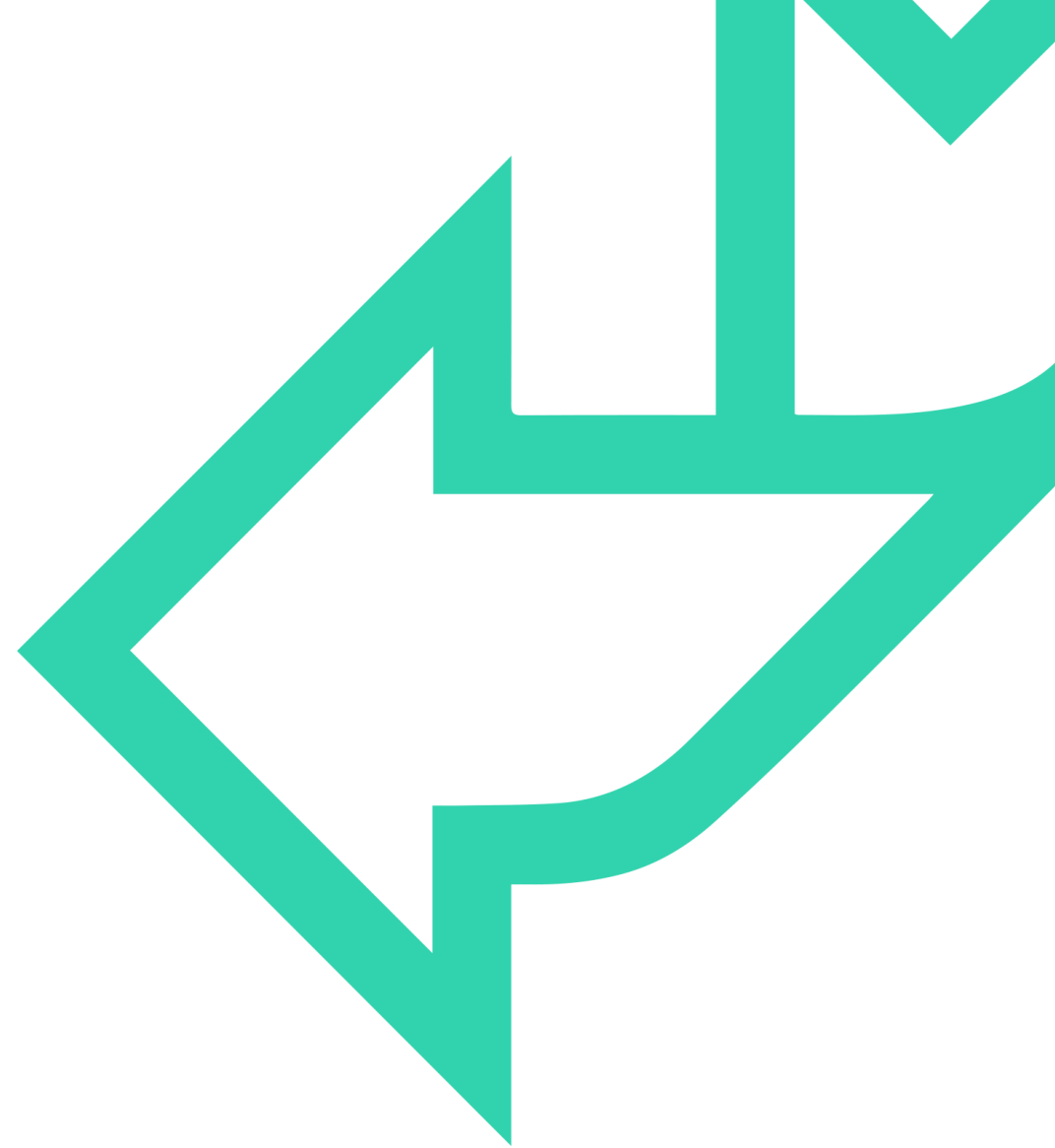




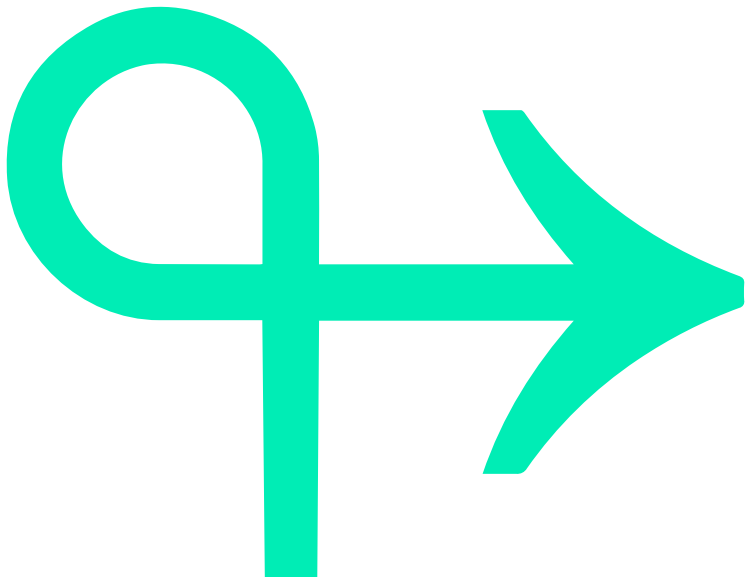
# **Module 9: Table Expressions**





# Table Expressions

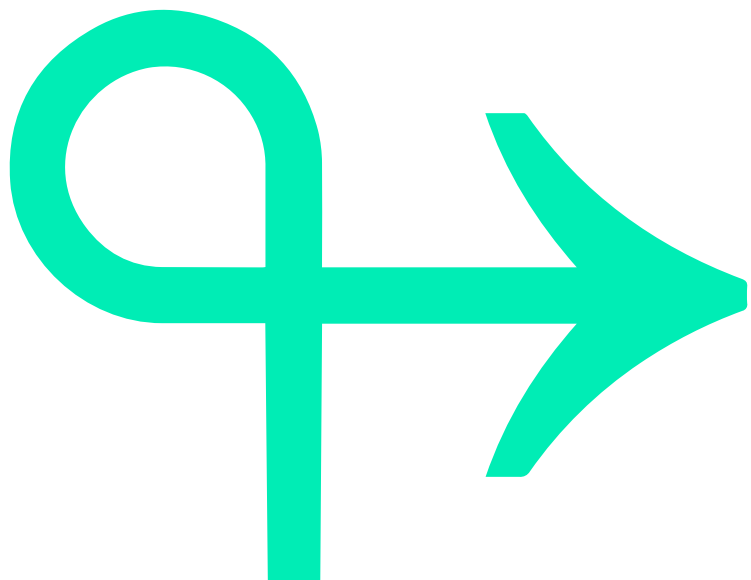
- Views
- Table-valued functions
- Derived tables
- Temporary tables
- Comparison





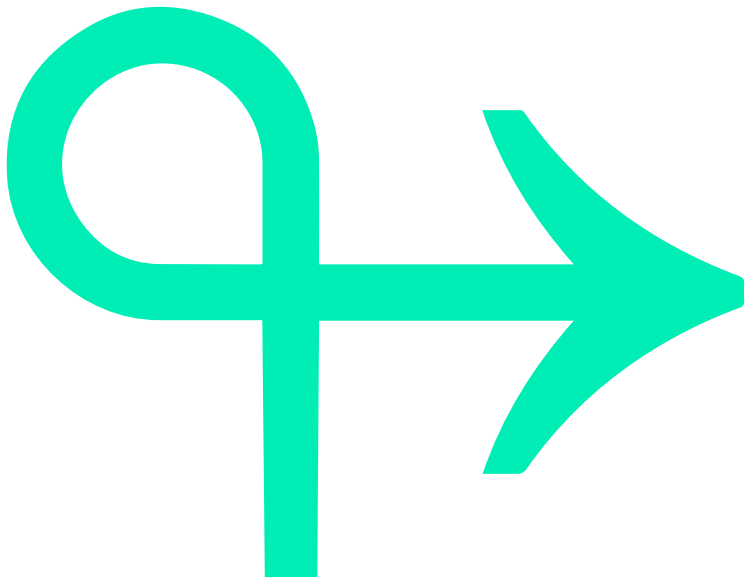
# Table Expressions

- Can be used to simplify the TSQL code by dividing the query into more easily understood parts.
- Temporary tables and table variables are not table expressions but can be used to perform the same purpose.





# Views



- Views are defined using a single SELECT statement.
- A view's definition is stored within the database for future use.
- A view can simplify the statements written by others where the same logic is reused.
- Administrators may use views to add security by not allowing access to the tables directly.
- ORDER BY is only permitted in a view if TOP, OFFSET/FETCH or FOR XML is used.

# QA CREATING VIEWS

Command outline:

```
CREATE VIEW <viewname> AS  
    SELECT <columns>  
    FROM <table(s) including joins>
```

Demonstration:

```
CREATE VIEW ProductCategories AS  
SELECT  
    Products.CategoryID,  
    CategoryName,  
    ProductID,  
    ProductName  
FROM  
    Products  
INNER JOIN  
    Categories  
ON  
    Products.CategoryID = Categories.CategoryID
```

Use:

```
SELECT * FROM ProductCategories
```



## **Module 9**

### **Exercise 1 – Views**

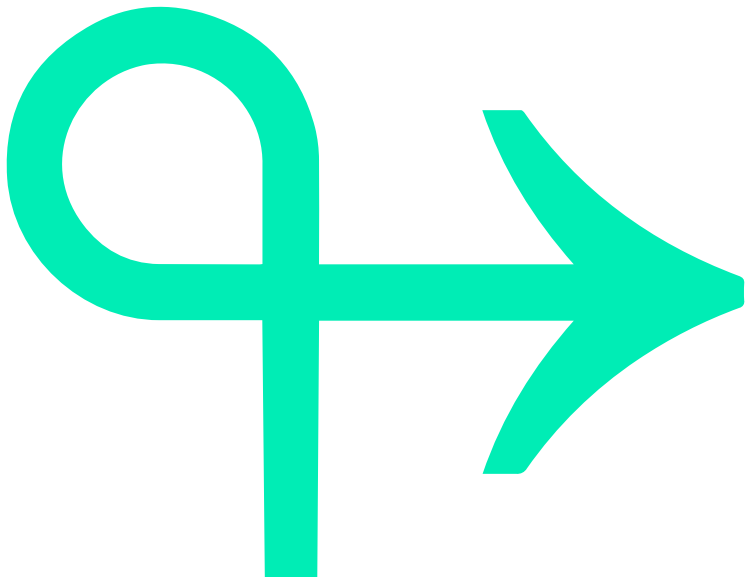
- Please complete exercise 1 of the Module 1 lab
- Stop at the end of task 2



# Table-Valued Functions (TVF)

## User defined table-valued functions

- The definition is stored within the database for others to use.
- Support input parameters.
- Used like a view.
- Use two-part naming convention when referring to objects such as tables within a function definition.





# Table-valued Functions (TVF)

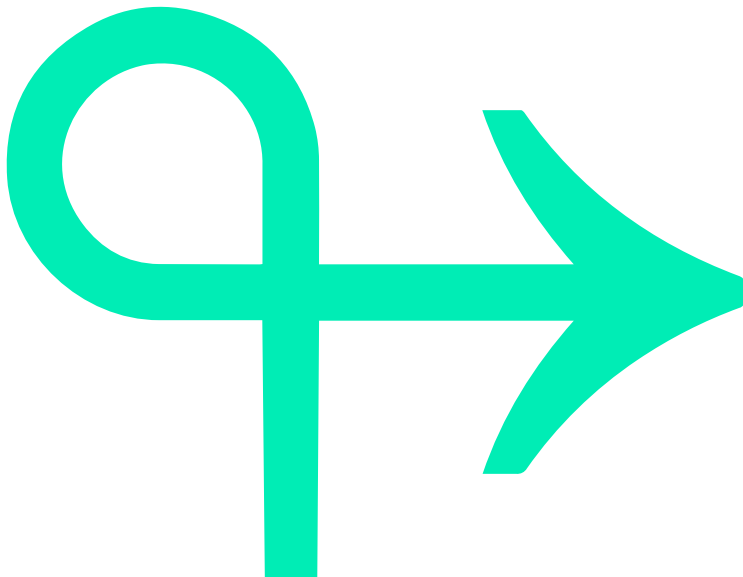
**Two types of user defined function can return tables.**

In-line:

- A single select statement.

Multi-line:

- Allows multiple lines for a more complex query.
- Table is pre-defined with the function and uses code to add rows.
- Use the create, alter and drop DDL statements.





# QA CREATING IN-LINE TVF

Command  
outline:

```
CREATE FUNCTION <functionname> (<parameters>)  
RETURNS TABLE AS  
    (SELECT <columns>  
    FROM <table(s) including joins>)
```

Demonstration:

```
CREATE FUNCTION GetProductsByCategory(@Category varchar(30))  
RETURNS TABLE AS  
RETURN(  
    SELECT  
        Categories.CategoryName,  
        Products.ProductName,  
        Products.QuantityPerUnit,  
        Products.UnitsInStock,  
        Products.Discontinued  
    FROM  
        Categories  
    INNER JOIN  
        Products  
    ON  
        Categories.CategoryID = Products.CategoryID  
    WHERE CategoryName = @Category  
)
```

Use:

```
SELECT * FROM GetProductsByCategory('Beverages')
```



# CREATING IN-LINE TVF

Use:

```
SELECT * FROM GetProductsByCategory('Beverages')  
-- returns all Beverages
```

```
SELECT * FROM GetProductsByCategory('Condiments')  
-- returns all Condiments
```

```
SELECT * FROM GetProductsByCategory('Wines')  
-- returns empty set – No wines are sold
```



## **Module 9**

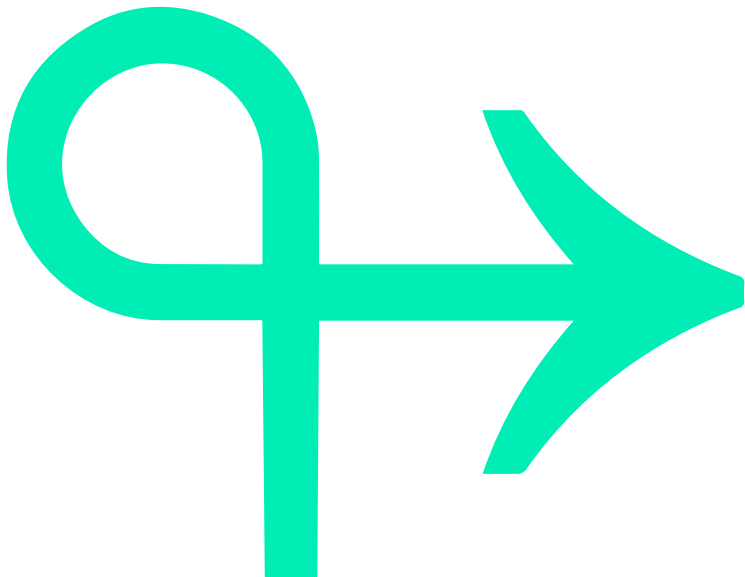
### **exercise 2 – in-line table valued functions**

- Please complete exercise 2 of the Module 1 lab
- Stop at the end of task 3



# Derived tables

- Derived tables are named query expressions created within an outer SELECT statement.
- Not stored in database – represents a virtual relational table.
- When processed, unpacked into query against underlying referenced objects.
- Allow you to write more modular queries.
- Scope of a derived table is the query in which it is defined.

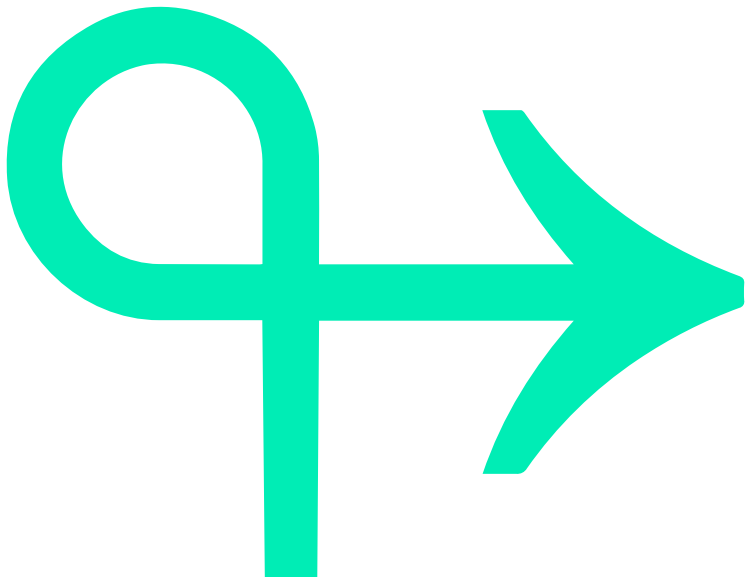




# DERIVED TABLE RULES

## Must:

- have an alias. AS
- have unique names for all columns.
- not use the ORDER BY clause unless in conjunction with TOP, OFFSET / FETCH, or FOR XML.
- use **From** (Select Query)



# QA DERIVED TABLE RULES

## Column names

- Defined as part of the inner query

```
SELECT *  
FROM  
(SELECT  
CategoryID,  
avg(UnitPrice) AS AvgPrice  
FROM  
Products  
GROUP BY  
CategoryID )  
AS  
Derivedtable
```

- Defined as part of the alias definition

```
SELECT *  
FROM  
(SELECT  
CategoryID,  
avg(UnitPrice) AS AvgPrice  
FROM  
Products  
GROUP BY  
CategoryID )  
AS  
Derivedtable (ProductCategoryID, AvgProductPrice)
```

# QA DERIVED TABLE EXAMPLE

- Show all products with above average list price for their subcategory

```
SELECT
P.CategoryID,
ProductName,
UnitPrice,
DT.AvgPrice
FROM
Products as P

INNER JOIN

(SELECT
categoryID,
avg(UnitPrice) AS AvgPrice
FROM
Products
GROUP BY
categoryID) AS DT
ON
P.categoryID = DT.categoryID
WHERE
UnitPrice >= DT.AvgPrice
```



## **Module 9**

### **exercise 3 – derived tables**

- Please complete exercise 3 of the Module 1 lab
- Stop at the end of task 2



# QA TEMPORARY TABLES

- Temporary tables can be used to store the results of a query for use later.
- Stored within TempDB (dropped when the user connection is dropped)
- Use **WITH**

## Can be:

- **local (#)** – only available to the creator, and is dropped automatically when the session ends.
- **global (##)** – available to all the users on the instance, and is dropped when all the users who have used the table finish their sessions.

## Commands:

- **CREATE** – creates the temporary table.
- **ALTER** – updates the design of the table.
- **DROP** – deletes the table.

# QA TEMPORARY TABLES

Command outline:

```
CREATE TABLE (# or ##)TableName(  
    column definitions  
)
```

```
INSERT INTO (# or ##)TableName  
    SELECT / VALUES to insert
```

```
DROP TABLE (# or ##)TableName
```

# QA TEMPORARY TABLES

Demonstration:

```
CREATE TABLE #Averages(  
    SubcategoryID int,  
    AverageListPrice money  
)  
GO  
  
INSERT INTO #Averages  
    SELECT ProductSubcategoryID, avg(ListPrice) AS AvgPrice  
        FROM Production.Product  
        GROUP BY ProductSubcategoryID  
GO  
  
SELECT *  
    FROM #Averages  
GO  
  
DROP TABLE #Averages
```



## **Module 9**

### **exercise 5 – temporary tables**

- Please complete exercise 5 of the Module 1 lab
- Stop at the end of task 3

# QA COMPARISON

|                                     | Views                                           | Table-valued function                                                                             | Derived tables                                          | Temporary table                                                           |
|-------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------|
| Main use                            | Storing the definition of a query for use later | Storing the definition of code for use later                                                      | Writing more complex queries than are possible normally | Storing a dataset for reuse later, by the session owner or other sessions |
| Definition stored inside database   | •                                               | •                                                                                                 |                                                         |                                                                           |
| Data exists                         | Single execution of the view                    | Single execution of the TVF                                                                       | Single execution of the query                           | Session                                                                   |
| Design shared with others           | •                                               | •                                                                                                 |                                                         | Depends on type                                                           |
| Recursive                           |                                                 |                                                                                                   |                                                         |                                                                           |
| Allow parameters to be passed       |                                                 | •                                                                                                 |                                                         |                                                                           |
| Allows access to declared variables |                                                 | Only declared variables within the function<br>Declared variables cannot be used with in-line TVF | •                                                       |                                                                           |



# Review

- **Views**
- **Table-valued functions**
- **Derived tables**
- **Temporary tables**
- **Comparison**

